

Range constructor for `std::span`

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1 Abstract

This paper proposes that `span` be constructible from any contiguous forwarding-range with a compatible element type. The idea was extracted from P1206.

2 Tony tables

Before	After
<pre>std::vector<int> v(42); std::span<int> foo = v view::take(3); <i>//ill-formed</i></pre>	<pre>std::vector<int> v(42); std::span foo = v view::take(3); <i>//valid</i></pre>
<pre>std::vector<int> v(42); std::span bar(v.begin(), 3); <i>// ill-formed</i></pre>	<pre>std::vector<int> v(42); std::span bar(v.begin(), 3); <i>// valid</i></pre>
<pre>std::vector<int> get_vector(); void foo(std::span<int>); void bar(std::span<const int>); bar(get_vector()); <i>//valid</i> foo(get_vector()); <i>//ill-formed</i></pre>	<pre>std::vector<int> get_vector(); void foo(std::span<int>); void bar(std::span<const int>); bar(get_vector()); <i>//valid</i> foo(get_vector()); <i>//ill-formed</i></pre>

3 Motivation

`std::span` is specified to be constructible from `Container` types. However, while defined, `Container` is not a concept and as such `ContiguousRange` is more expressive. Furthermore, there exist some non-container ranges that would otherwise be valid ranges to construct `span` from. As such `span` as currently specified fits poorly with the iterators / ranges model of the rest of the standard library.

The intent of `span` was always to be constructible from a wide number of compatible types, whether standard contiguous containers, non-standard equivalent types, or views. This proposal ensure that

span, especially when used as parameter of a function will be constructible from all compatible types while offering stronger and more consistent (in regard to Range) lifetime guarantees.

4 Design considerations

We propose to specify all constructors currently accepting a container or pointers in terms of `ContiguousRange` and `ContiguousIterator` respectively as well as to add or modify the relevant deduction guides for these constructors.

5 Future work

- We suggest that both the wording and the implementation of span would greatly benefit from a trait to detect whether a type has a static extent. Because `std::extent` equals to 0 for types without static extent, and because 0 is a valid extent for containers, `std::extent` proved too limited. However we do not propose a solution in the present paper.

6 Proposed wording

Change in **[views.span]** 21.7.3:

```
// [span.cons], constructors, copy, and assignment
constexpr span() noexcept;
template <class It>
constexpr span( pointer_ptr It begin, index_type count);
constexpr span(pointer first, pointer last);
template <class It, class End>
constexpr span(It first, End last);

template<size_t N>
constexpr span(element_type (&arr)[N]) noexcept;
template<size_t N>
constexpr span(array<value_type, N>& arr) noexcept;
template<size_t N>
constexpr span(const array<value_type, N>& arr) noexcept;
template<class Container>
constexpr span(Container& cont);
template<class Container>
constexpr span(const Container& cont);
template <class R>
constexpr span(R&& r);

constexpr span(const span& other) noexcept = default;
template<class OtherElementType, ptrdiff_t OtherExtent>
constexpr span(const span<OtherElementType, OtherExtent>& s) noexcept;
```

...

}

```
template<class T, size_t N>
span(T (&)[N]) -> span<T, N>;
template<class T, size_t N>
span(array<T, N>&) -> span<T, N>;
template<class T, size_t N>
span(const array<T, N>&) -> span<const T, N>;
template <class It, class End>
span(It, End) -> span<remove_reference_t<iter_reference_t<It>>>>;

template<class T, size_t N>
span(const array<T, N>&) -> span<const T, N>;
template<class Container>
span(Container&) -> span<typename Container::value_type>;
template<class Container>
span(const Container&) -> span<const typename Container::value_type>;
template<class R>
span(R&&) -> span<ranges::range_value_t<R>>;
```

In 21.7.3.2 [span.cons]

constexpr span() noexcept;

Ensures: size() == 0 && data() == nullptr.

Remarks: This constructor shall not participate in overload resolution unless Extent <= 0 is true.

constexpr span(pointer ptr, index_type count);

template <class It>
constexpr span(It first, index_type count);

Constraints:

- It satisfies ContiguousIterator,
- is_convertible_v<remove_reference_t<iter_reference_t<It>>(*)[], element_type(*)[]> is true. [*Note:* The intent is to allow only qualification conversions of the iterator reference type to element_type — end note]

Expects:

- [~~ptr~~ first, ~~ptr~~ first + count) shall be a valid range.
- It models ContiguousIterator,
- If extent is not equal to dynamic_extent, then count shall be equal to extent.

Effects: Constructs a `span` that is a view over the range `[ptr first , ptr first + count)`.

Ensures: `size() == count && data() == ptr`.

- `size() == count` is true
- `to_address(first) == data()` is true

Throws: Nothing.

```
constexpr span(pointer first, pointer last);
```

Requires: `[first, last)` shall be a valid range. If `extent` is not equal to `dynamic_extent`, then `last - first` shall be equal to `extent`.

Effects: Constructs a span that is a view over the range `[first, last)`.

Ensures: `size() == last - first && data() == first`.

Throws: Nothing.

```
template <class It, class End>  
constexpr span(It first, End last);
```

Constraints:

- `is_convertible_v<remove_reference_t<iter_reference_t<It>>(*), element_type(*)[]>` is true, [*Note:* The intent is to allow only qualification conversions of the iterator reference type to `element_type` — *end note*] ,
- It satisfies `ContiguousIterator`,
- End satisfies `SizedSentinel<It>`,
- `is_convertible_v<End, size_t>` is false.

Expects:

- If `extent` is not equal to `dynamic_extent`, then `last - first` is equal to `extent`,
- `[first, last)` is a valid range,
- It models `ContiguousIterator`,
- End models `SizedSentinel<It>`.

Effects: Constructs a span that is a view over the range `[first, last)`.

Ensures:

- `size() == last - first` is true,
- `to_address(first) == data()` is true.

Throws: Nothing.

```
template<size_t N> constexpr span(element_type (&arr)[N]) noexcept;
template<size_t N> constexpr span(array<value_type, N>& arr) noexcept;
template<size_t N> constexpr span(const array<value_type, N>& arr) noexcept;
```

Effects: Constructs a `span` that is a view over the supplied array.

Ensures: `size() == N && data() == data(arr)`.

Remarks: These constructors shall not participate in overload resolution unless:

- `extent == dynamic_extent || N == extent` is true, and
- `remove_pointer_t<decltype(data(arr))>(*) []` is convertible to `element_type(*) []`.

```
template<class Container> constexpr span(Container& cont);
template<class Container> constexpr span(const Container& cont);
```

Constraints:

- `extent == dynamic_extent` is true,
- `Container` is not a specialization of `span`,
- `Container` is not a specialization of `array`,
- `is_array_v<Container>` is false,
- `data(cont)` and `size(cont)` are both well-formed, and
- `remove_pointer_t<decltype(data(cont))>(*) []` is convertible to `ElementType(*) []`.

Expects: `[data(cont), data(cont) + size(cont))` is a valid range.

Effects: Constructs a `span` that is a view over the range `[data(cont), data(cont) + size(cont))`.

Ensures: `size() == size(cont) && data() == data(cont)`.

Throws: What and when `data(cont)` and `size(cont)` throw.

```
template <class R>
constexpr span(R&& r);
```

Constraints:

- `extent == dynamic_extent` is true,
- `R` satisfies `ranges::ContiguousRange` and `ranges::SizedRange`,
- either `R` satisfies *forwarding-range* or `is_const_v<element_type>` is true,
- `remove_reference_t<R>` is not a specialization of `span`,
- `remove_reference_t<R>` is not a specialization of `array`,
- `is_array_v<remove_reference_t<R>>` is false,

- `is_convertible_v<remove_reference_t<iter_reference_t<ranges::iterator_t<R>>>(*), element_type(*)>>` is true [*Note*: The intent is to allow only qualification conversions of the iterator reference type to `element_type` — *end note*] .

Expects: `R` models `ranges::ContiguousRange` and `ranges::SizedRange`.

Ensures: `size() == ranges::size(r) && data() == ranges::data(r)` is true.

Throws: What and when `ranges::data(r)` and `ranges::size(r)` throw.

```
constexpr span(const span& other) noexcept = default;
```

Ensures: `other.size() == size() && other.data() == data()`.

Add a new section [span.deduction] to describe the following deduction guides:

```
template <class It, class End>
span(It, End) -> span(It, End) -> span<remove_reference_t<iter_reference_t<It>>>>;
```

Constraints:

- `It` satisfies `ContiguousIterator`,
- `End` satisfies `SizedSentinel<It>`.

```
template<class R>
span(R&&) -> span<ranges::range_value_t<R>>;
```

Constraints: `R` satisfies `ranges::ContiguousRange`.

7 References

- [P1419] Casey Carter, Corentin Jabot *A SFINAE-friendly trait to determine the extent of statically sized containers*
<https://wg21.link/P1419>
- [P1391] Corentin Jabot *Range constructor for std::string_view*
<https://wg21.link/P1391>
- [P1474] Casey Carter *Helpful pointers for ContiguousIterator*
<https://wg21.link/P1474>